

What is Array?

- Arrays in Java are powerful tools that allow you to store multiple elements of the same data type in a single variable.
- They provide a structured and efficient way to organize and manipulate data.

Declaration



```
//syntax  
datatype [] arrayName;  
//example  
int[] arr;
```

Initialization



```
//Syntax  
datatype [] arrayName = new datatype [ size ];  
//example  
int[] arr = new int[20];
```

Types of Array

- **Single Dimensional Array**

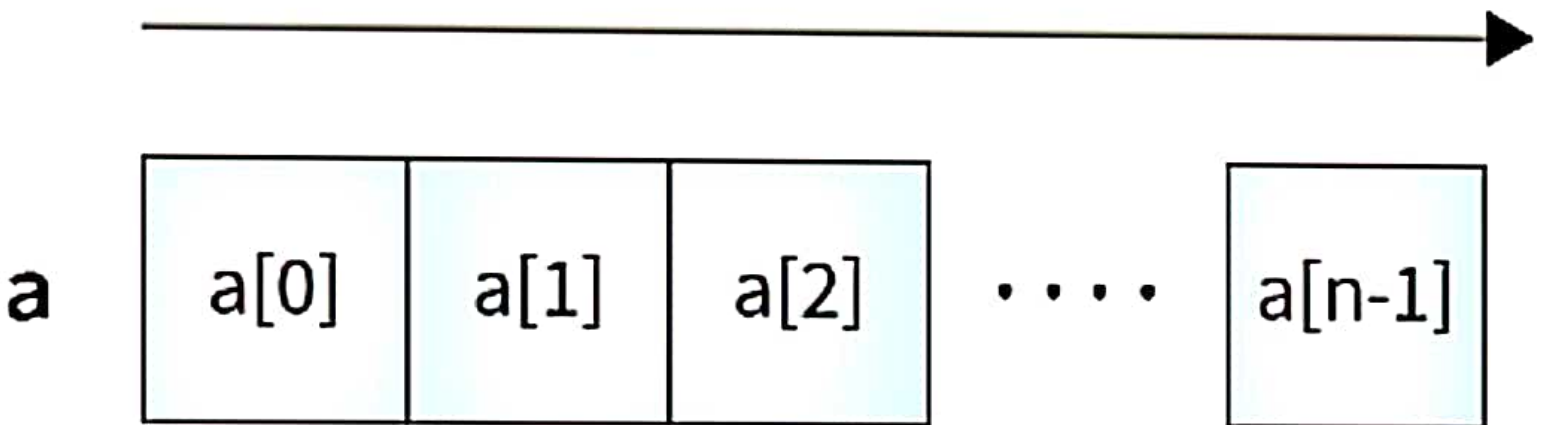
An array that has only one subscript or one dimension is known as a single-dimensional array.

- **Multi-Dimensional Array**

A multi-dimensional array is just an array of arrays that represents multiple rows and columns.

Single Dimensional Array

One-Dimension

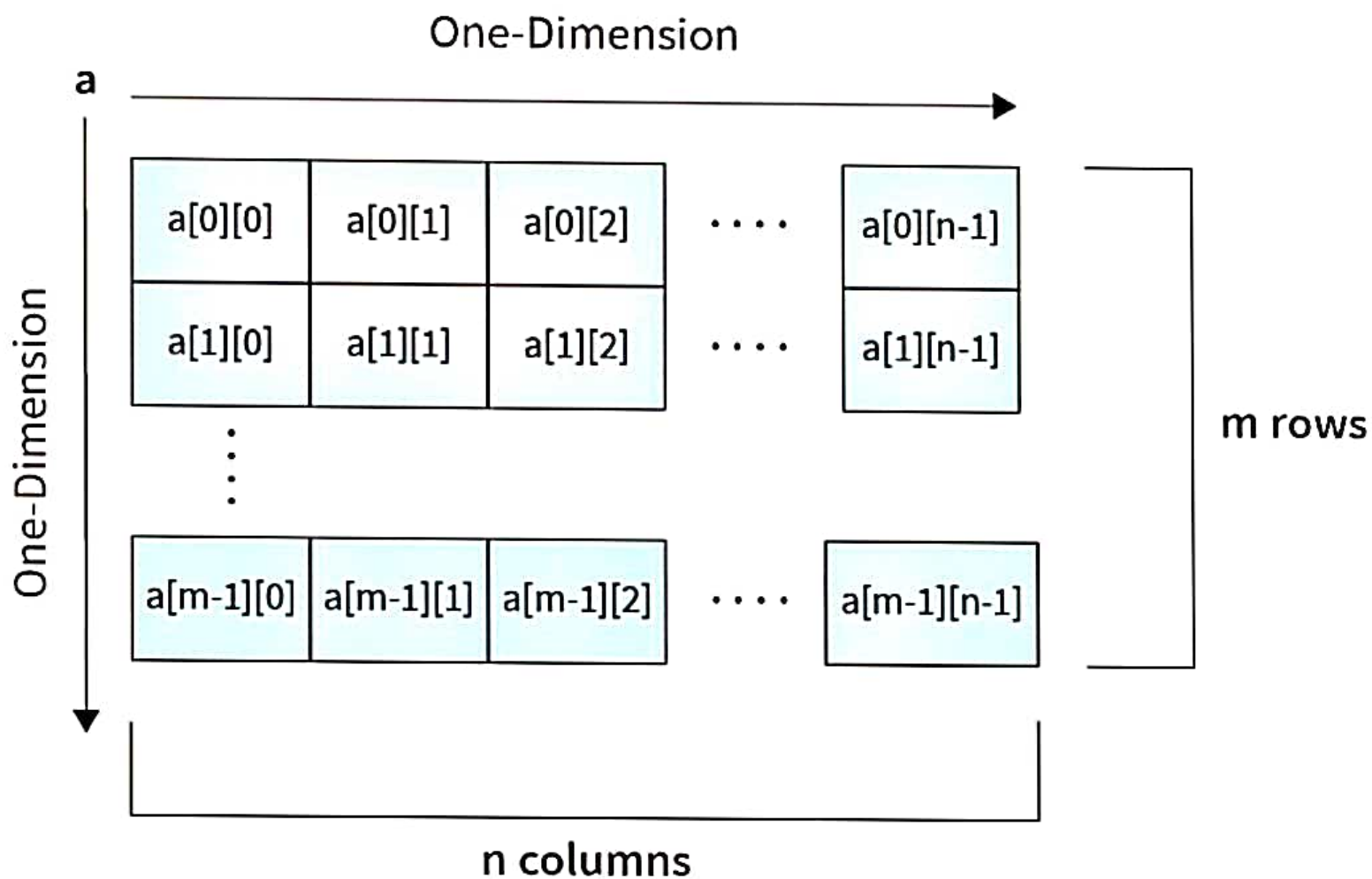


Example



```
int marks[];           // declared an array
marks = new int[5];    // allocating memory
// initializing array using index
marks[0] = 56;
marks[1] = 98;
marks[2] = 77;
marks[3] = 89;
marks[4] = 99;
```

Multi-Dimensional Array



Example



```
int marks[][] = {  
    {77, 85, 68, 99, 87},  
    {98, 56, 79, 90, 92},  
    {78, 88, 56, 70, 99}  
};
```

//OR

```
int marks[][] = new int[3][5];  
// both will create a 2D array  
//with 3 rows and 5 columns.
```